CLASSROOM CONTACT PROGRAMME
(Academic Session : 2025 - 2026)

LEADER & ACHIEVER COURSE

PHASE : ALL SRG ENTHUSE, LEADER, ACHIEVER COURSE & LAKSHYA PHASE

TARGET : PRE-MEDICAL 2026

Test Type :**SRG-MAJOR** Test Pattern :**NEET (UG)**

TEST DATE : 13-06-2026

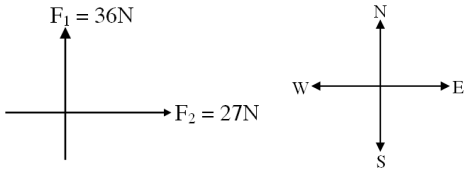
ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	2	4	1	4	4	2	3	1	3	2	2	3	1	2	4	2	1	1	4	3	4	3	2	1	3	2	2	1	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	2	1	4	4	4	4	4	2	1	3	1	2	4	3	1	1	2	1	2	2	3	2	4	4	4	4	1	3	1	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	1	3	3	4	1	4	1	4	2	3	4	3	3	3	3	1	1	1	2	2	2	2	4	4	4	4	4	3	4	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	4	2	4	3	1	3	4	1	4	2	1	2	4	3	4	4	3	3	4	3	2	4	3	2	1	3	3	4	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	1	2	2	2	1	3	4	2	1	2	2	4	2	2	3	2	3	1	2	4	2	1	3	3	2	1	4	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	4	4	2	3	3	2	2	3	4	4	1	3	1	2	1	3	2	1	3	1	4	2	4	1	4	3	3	3	2	1

HINT – SHEET

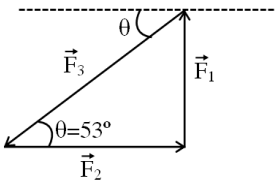
1. Ans (3)

According to question



and $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = 0$ for equilibrium

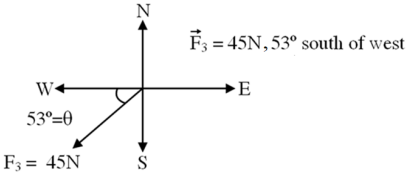
also, by Pythagorus theorem



we can say

$F_3 = \sqrt{F_1^2 + F_2^2} = \sqrt{27^2 + 36^2} = 45$ N

$\tan\theta = \frac{F_1}{F_2} = \frac{36}{27} = \frac{4}{3} \Rightarrow \theta = 53°$



2. Ans (2)

KE per unit mass = $\frac{V^2}{2}$ dimension (L^2T^{-2})

$\frac{V^2}{2} = \alpha\gamma^x\sigma^y r^z$

$L^2T^{-2} = (ML^{-3})^x(M^1L^{-1}T^{-2})^yL^2$

$M^0L^2T^{-2} = M^{x+y}L^{-3x-y+z}T^{-2y}$

$x + y = 0 \quad -3x - y + z = 2 \quad -2 = -2y$

solving above

$y = 1 \; ; x = -1 \; ; 3 - 1 + z = 2 \Rightarrow z = 0$

$x + y + z = 0$

3. Ans (4)

(1) Energy, Power, Frequency :

Power = Energy × Frequency

Since they are dependent, they cannot be taken as fundamental.

(2) Momentum, Velocity, Energy :

Dimensionally [E] = [p][v]

Since energy can be expressed in terms of momentum and velocity, this set cannot enter into list of fundamental PQ's.

(3) Length, Acceleration, Time :

Since $[\text{Acceleration}] = \frac{[\text{Length}]}{[\text{Time}]^2}$, these are dependent and cannot enter into fundamental PQ's.

(4) Mass, Acceleration, Velocity :

Since in any of these, one cannot be expressed in terms of others, so they can be taken as fundamental PQ.

Alternate : Determinant of dimension should be non zero to be fundamental quantity. Determinant of option (4) is non zero, but zero for (1), (2) & (3).

$$\begin{array}{c} \\ \text{Mass} \\ \text{Acceleration} \\ \text{Velocity} \end{array}\begin{bmatrix} M & L & T \\ 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 1 & -1 \end{bmatrix} = 1\,(-1+2) = 1$$

4. Ans (1)

$$R = \rho\frac{\ell}{A} = \frac{V}{I}$$

$$\Rightarrow \rho = \frac{AV}{I\ell} = \frac{\pi d^2 V}{4I\ell}$$

$$\therefore \frac{\Delta\rho}{\rho} = \frac{2\Delta d}{d} + \frac{\Delta V}{V} + \frac{\Delta I}{I} + \frac{\Delta\ell}{\ell}$$

$$\frac{\Delta\rho}{\rho} = 2\left(\frac{0.01}{5.00}\right) + \frac{0.1}{5.0} + \frac{0.01}{2.00} + \frac{0.1}{10.0}$$

$$\frac{\Delta\rho}{\rho} = 0.004 + 0.02 + 0.005 + 0.01$$

$$\frac{\Delta\rho}{\rho} = 0.039$$

$$\%\ \text{error} = \frac{\Delta\rho}{\rho} \times 100 = 0.039 \times 100 = 3.90\%$$

5. Ans (4)

The least count is determined from the main scale and vernier scale divisions. The zero error is found from the given information when the jaws are touching. The corrected reading is the observed reading minus the zero error.

$T = \frac{30\,\text{sec}}{20}$ $\quad \Delta T = \frac{1}{20}$ sec.

$L = 55$ cm $\quad \Delta L = 1\text{mm} = 0.1$ cm

$$g = \frac{4\pi^2 L}{T^2}$$

percentage error in g is

$$\frac{\Delta g}{g} \times 100\% = \left(\frac{\Delta L}{L} + \frac{2\Delta T}{T}\right) 100\%$$

$$= \left(\frac{0.1}{55} + \frac{2\left(\frac{1}{20}\right)}{\frac{30}{20}}\right) 100\% \simeq 6.8\%$$

6. Ans (4)

$$Y = \frac{mg}{A}\frac{\ell}{\Delta\ell} = \frac{Mg}{\pi r^2}\frac{\ell}{\Delta\ell}$$

$$\frac{\Delta Y}{Y} \times 100 = \left[\frac{2\Delta r}{r} + \frac{\Delta\ell}{\ell} + \left[\Delta\frac{(\Delta\ell)}{\Delta\ell}\right]\right] \times 100\%$$

$$= \left[2 \times \frac{0.001}{0.05} + \frac{0.1}{100} + \frac{0.001}{0.125}\right] \times 100\%$$

$$= \left[\frac{2}{50} + \frac{1}{1000} + \frac{1}{125}\right] \times 100$$

%

$$= [0.04 + 0.001 + 0.008] \times 100 \approx 4.9\ \%$$

7. Ans (2)

$$\frac{R_x}{R} = \frac{x}{L-x} \Rightarrow R_x = R\left(\frac{x}{L-x}\right)$$

For $R = 20\Omega$, $x = 20$ cm

$$\Rightarrow R_x = 20\left[\frac{20}{100-20}\right] = 5\Omega$$

from (i) $\ell n R_x = \ell n R + \ell n x - \ell n\,(L-x)$

$$\frac{dR_x}{R_x} = \frac{dx}{x} + \frac{dx}{L-x}$$

$$\frac{\Delta R_x}{R_x} = \frac{L\Delta x}{x(L-x)}\ [\Delta x = \text{maximum error in x}]$$

for $R_x = 5\Omega$; $|\Delta x| = 0.1$ cm;

$L = 100$ cm; $x = 20$ cm

$$\therefore |\Delta R_x| = \frac{5 \times 100 \times 0.1}{20 \times 80} = 0.03\Omega$$

$$\therefore R_x = (5.00 \pm 0.03)\Omega$$

8. Ans (3)

AB = x

BC = x

AD = $2x$ + CD = $3x$

CD = x

$$<v> = \frac{\text{total distance}}{\text{total time}}$$

$$< v >= \frac{3x}{\frac{x}{v_1} + \frac{x}{v_2} + \frac{x}{v_3}} = \frac{3v_1v_2v_3}{v_2v_3 + v_1v_3 + v_1v_2}$$

9. Ans (1)

Vertical co-ordinates at time t :

• For-1 : $y_1 = u_1 \sin t - \frac{1}{2}gt^2$.

• For-1 : $y_2 = u_2 t - \frac{1}{2}gt^2$.

For collision $y_1 = y_2$.

$u_1 \sin\theta t = u_2 t$.

For $t \neq 0$, $u_1 \sin\theta = u_2 \Rightarrow \frac{u_1}{u_2} = \frac{1}{\sin\theta}$

With $\theta = 60^o$, $\sin 60^o = \frac{\sqrt{3}}{2}$ so,

$$\frac{u_1}{u_2} = \frac{1}{\sqrt{3}/2} = \frac{2}{\sqrt{3}}$$

10. Ans (3)

With respect to river frame both will collide at some point on AB line. Now come to ground frame by adding displacement $(\vec{v}_{river})$ t which will bring the colliding point east of line AB.

11. Ans (2)

P will increase till t = 5, then P decrease

P is maximum at t = 5

$\int F\,dt = \Delta P$

$10 \times 5 = m\,(v_f - v_i)$

$50 = 2\,(v - 2)$

v = 27 m/s ⟶ maximum speed

12. Ans (2)

Tension = T − mg = ma

T = m(g + a)

T = 10(10 + 2)

T = 10 × 12 = 120

Upper support, 120 + 80 = 200

(80) weight of latter monkey

13. Ans (3)

As slope increases from O to A ⇒ velocity increases

As slope is constant from A to B ⇒ velocity constant

As slope decreases from B to C ⇒ velocity decreases

Work done = Δk

As velocity increases ⇒ Δk increases ⇒ and vice a versa.

14. Ans (1)

$$\frac{V^2}{R} = kRt$$

$V^2 = kR^2t$

$$V = \sqrt{kR^2}\sqrt{t}$$

$$\frac{dv}{dt} = \sqrt{kR^2} \times \frac{1}{2\sqrt{t}}$$

$$= \frac{\sqrt{kR^2}}{2\sqrt{t}}$$

$F = Ma_t$

$$= \frac{M}{2} \times \sqrt{\frac{kR^2}{t}}$$

$P = F.V$

$$= \frac{M}{2} \times \sqrt{\frac{kR^2}{t}} \times \sqrt{kR^2t}$$

$$= \frac{MkR^2}{2}$$

15. Ans (2)

$j = \sigma E = nev_d$

$$\sigma = ne\frac{v_d}{E}$$

$= ne\mu$

$$\frac{1}{\sigma} = \rho = \frac{1}{n_e e \mu_e}$$

$$= \frac{1}{10^{19} \times 1.6 \times 10^{-19} \times 1.6} = 0.4\ \Omega m$$

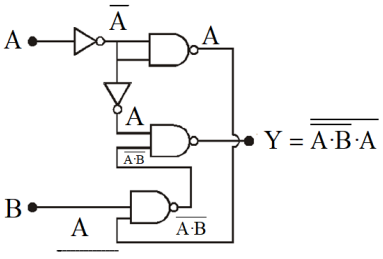
16. Ans (4)

Both connecting a capacitor in shunt and an inductor in series help to smooth the pulsating DC output (reduce ripple).

- **Statement I (Capacitor Shunt):** True.
- **Statement II (Inductor Series):** True.

To convert pulsating dc into steady dc both of mentioned method are correct.

17. Ans (2)



$Y = \overline{AB}.A$

$= \overline{AB} + \overline{A}$

$= 0 + 0$

$= 0$

18. Ans (1)

(A) Nuclear fusion: This is the process where two light nuclei combine to form a heavier nucleus and the energy is released because some of the matter is converted into energy.

(B) Nuclear fission: This is the process where a heavy nucleus splits into two or more smaller nuclei. It generally occurs for nuclei with higher atomic number (like Uranium or Plutonium). The energy is released because some of the matter is converted into energy.

(C) Beta decay : This is a type of radioactive decay that involves the conversion of a neutron into a proton or a proton into a neutron, accompanied by the emission of an electron or a positron, and a antineutrino or neutrino. This process is mediated by the weak nuclear force. The energy is released because some of the matter is converted into energy.

(D) Exothermic nuclear reaction: An exothermic nuclear reaction converts some matter into energy. Both fusion and fission are examples of highly exothermic Nuclear Reactions.

19. Ans (1)

Threshold wavelength $\lambda_{th} = 240$ nm

$\therefore$ Work function, $\phi = \dfrac{hc}{\lambda_{th}}$

$= \dfrac{(4 \times 10^{-15}\text{eV s})(3 \times 10^{8}\text{ms}^{-1})}{240 \times 10^{-9}} = 5\,\text{eV}$

With light of wavelength $2\lambda_0$, the maximum kinetic energy of emitted electrons is 10 eV.

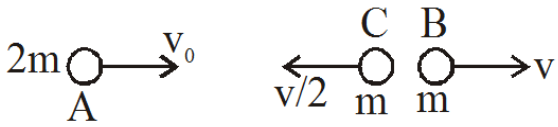
$\therefore$ Energy of photon of wavelength $2\lambda_0$ is $= 10 + 5 = 15$ eV

$\therefore$ Energy of photon of wavelength λ_0 is $= 15 \times 2 = 30$ eV

With light of wavelength λ_0, the maximum kinetic energy of emitted electrons is $= 30 - 5 = 25$ eV

$\therefore$ $V_0 = 25$ V

20. Ans (4)



let mass of B and C is m each.

By momentum conservation

$2mv_0 = mv - \dfrac{mv}{2}$

$p_A = 2mv_0 \quad p_B = 4mv_0, \; p_c = 2mv_0$

De-Broglie wavelength $\lambda = \dfrac{h}{p}$

$\lambda_A = \dfrac{h}{2mv_0}; \lambda_B = \dfrac{h}{4mv_0}; \lambda_C = \dfrac{h}{2mv_0}$

21. Ans (3)

$\Delta Q = mS\Delta T$

$\left(\Delta T = \dfrac{1}{ms}\Delta Q\right)$

slope of $T - Q$ curve $= \dfrac{1}{ms}$

$mL_V = 200$

$mL_f = 100$

22. Ans (4)

$\eta = \left(1 - \dfrac{Q_2}{Q_1}\right) \times 100 = \left(1 - \dfrac{1500}{2000}\right) \times 100$

$= \left(1 - \dfrac{3}{4}\right) \times 100$

$\eta = (1 - 0.75) \times 100$

$\eta = 0.25 \times 100$

$\eta = 25\%$

23. Ans (3)

$\frac{\partial y}{\partial t} = 2A\omega \sin\left(\frac{\pi x}{L}\right)\cos\omega t$

For antinode (at $x = \pm = \frac{L}{2}$)

$\frac{\partial y}{\partial t} = 2A\omega\cos\omega t$

now for the given particle

$\frac{\partial y}{\partial t} = 2A\omega \sin\left(\frac{\pi x}{L}\right) = A\omega$

$\therefore \sin\frac{\pi x}{L} = \frac{1}{2}$

$\therefore x = \frac{L}{6}$

Hence : answer is $\frac{L}{3}$

24. Ans (2)

$V_1\rho_1 \times g + V_2\rho_2 \times g = (V_1 + V_2)\rho g$

$V_1\rho_1 + V_2\rho_2 = V_1\rho + V_2\rho$ .... (i)

$V_1 + V_2 = V$ .... (ii)

$\frac{V_1}{V} = \frac{\rho_2 - \rho}{\rho_2 - \rho_1}$

25. Ans (1)

$I_1 = \frac{m\ell^2}{3} + m\ell^2 = \frac{4}{3}m\ell^2$

$I_2 = \frac{m\ell^2}{3}$

$I_3 = \frac{m\ell^2}{12}$

$I_4 = \frac{m\ell^2}{3} + m\left(\frac{\ell}{2}\right)^2 = \frac{m\ell^2}{3} + \frac{m\ell^2}{4}$

$I_4 = \frac{7m\ell^2}{12}$

$I_1 > I_4 > I_2 > I_3$

26. Ans (3)

$V = -\int \vec{E}\cdot d\vec{r}$

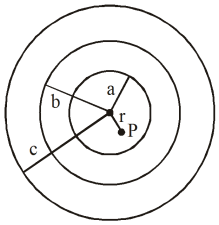
$V = -\int \left(\sqrt{3}\hat{i} - \hat{j}\right)\cdot\left(dx\,\hat{i} + dy\,\hat{j}\right)$

$V = -\left[\sqrt{3}x - y\right]$

$V = -\sqrt{3}x + y$

So, equipotential line having slope 60°

27. Ans (2)



Potential at point P, $V = \frac{kQ_a}{a} + \frac{kQ_b}{b} + \frac{kQ_c}{c}$

$\because Q_a : Q_b : Q_c :: a^2 : b^2 : c^2$

{since $\sigma_a = \sigma_b = \sigma_c$}

$\therefore Q_a = \left[\frac{a^2}{a^2 + b^2 + c^2}\right]Q$

$Q_b = \left[\frac{b^2}{a^2 + b^2 + c^2}\right]Q$

$Q_c = \left[\frac{c^2}{a^2 + b^2 + c^2}\right]Q$

$V = \frac{Q}{4\pi\epsilon_0}\left[\frac{(a+b+c)}{a^2 + b^2 + c^2}\right]$

28. Ans (2)

Direction of Electric field. at B is towards A that will exert force in this direction only, causing the positive charge to move [$\vec{E}$ is perpendicular to equipotential surface and its direction is from high potential to low potential]

29. Ans (1)

$\vec{P} = 30 \times 10^{-5}$Cm

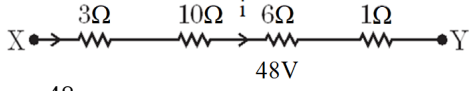
Using Gauss law

$\phi = \frac{Q_{in}}{\varepsilon_0}$ and $Q_{in} = 0$

$\Rightarrow \phi = 0$

Assertion and Reason are correct.

30. Ans (1)



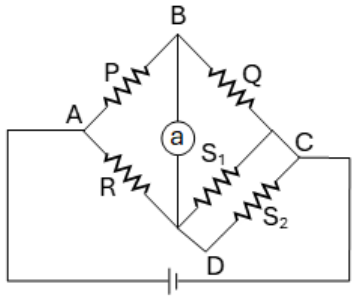
$i = \frac{48}{6} = 8$

$R_{net} = 20\Omega$

$V_{xy} = 160$ volt

31. Ans (2)

For balanced Wheatstone bridge.



$$\frac{P}{Q} = \frac{R}{\frac{S_1S_2}{S_1+S_2}} = \frac{R(S_1+S_2)}{S_1S_2}$$

32. Ans (1)

$$B = \frac{\mu_0}{4\pi} q \frac{vr}{r^3} = \frac{\mu_0 q\omega}{4\pi r}$$

$$14 = \frac{10^{-7} \times 1.6 \times 10^{-19} \times \omega}{0.5 \times 10^{-10}}$$

$$\omega = \frac{7}{1.6} \times 10^{16} \text{rad/s}$$

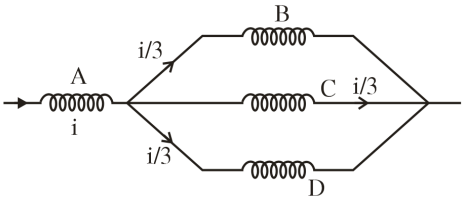
33. Ans (4)

These (A,C only) are the property of paramagnetic material.

34. Ans (4)

The area of B-H Hysteresis loop represents the energy dissipated per cycle per unit volume.

35. Ans (4)



$\phi \propto i$

$\Rightarrow B \propto i$

so, field at centre of C $= \frac{3}{3} = 1T$

36. Ans (4)

$$I_{rms} = \sqrt{\frac{\int i^2 dt}{\int dt}}$$

$$i = \frac{2i_0}{T}t$$

$$\therefore I_{rms} = \left[\frac{\frac{4i_0^2}{T^2}\left[\frac{t^3}{3}\right]_0^{T/2}}{\frac{T}{2}}\right]^{1/2} = \frac{i_0}{\sqrt{3}}$$

37. Ans (4)

Amplitude of electric field

$E_0 = B_0C = 60$ V/m

direction of E $= -\hat{j}$

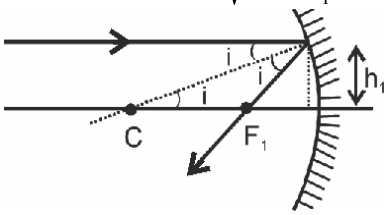
38. Ans (2)

Since Apparent depth $= \frac{\text{Real depth}}{\mu}$

Length of Image $= L + \frac{L}{\mu}$

39. Ans (1)

$$CF_1 = \frac{R}{2\cos i} = \frac{R^2}{2\sqrt{R^2 - h_1^2}} (h_1 = 3 \text{ cm})$$



$$CF_2 = \frac{R^2}{2\sqrt{R^2 - h_2^2}} \quad (h_2 = 4 \text{ cm})$$

$$\Rightarrow CF_2 - CF_1 = \frac{25}{24} \text{ cm}$$

40. Ans (3)

Here, m = 30, $f_e = 5$ cm, D = 25 cm

Magnifying power of a compound microscope is

$$m = m_0 \times m_e = m_0\left(1 + \frac{D}{f_e}\right)$$

or $30 = m_0\left(1 + \frac{25}{5}\right)$ or $m_0 = 5$

41. Ans (1)

$n_1\lambda_1 = n_2\lambda_2$

$60 \times 4000 = n_2 \times 6000$

$n_2 = 40$

42. Ans (2)

$y = a_0 \sin(\omega t - kx)$

at $x = \frac{\pi}{2k}$ at $t = 0$,

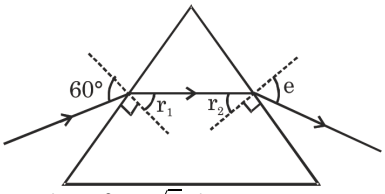
$y = a_0 \sin[\omega \times 0 - 2\pi \times \frac{\pi}{2k}]$

$y = a_0 \sin[-\frac{\pi}{2}] = -a_0 \sin\frac{\pi}{2}$

$y = -a_0$

So, point Q is at max(−ve) displacement.

43. Ans (4)



$1 \cdot \sin 60^o = \sqrt{3} \sin r_1$

$r_1 = 30^o$

$r_2 = A - r_1$

$= 30^o$

$\sqrt{3} \sin r_2 = 1 \cdot \sin e$

$e = 60^o$

44. Ans (3)

- Frequency formula: $f = \frac{1}{LD}\sqrt{\frac{T}{\pi\rho}}$
- Frequency ratio: $\frac{f_1}{f_2} = \frac{L_2}{L_1} \cdot \frac{D_2}{D_1}\sqrt{\frac{T_1}{T_2} \cdot \frac{\rho_2}{\rho_1}}$
- Substituting ratios:

$\frac{f_1}{f_2} = \frac{35}{36} \times \frac{1}{4} \times \sqrt{\frac{8}{1} \times \frac{2}{1}} = \frac{35}{36} \times \frac{1}{4} \times 4 = \frac{35}{36}$

- Given higher frequency $(f_2) = 360$ Hz
- Lower frequency $(f_1) = \frac{35}{36} \times 360 = 350$ Hz
- Beat frequency = $f_2 - f_1 = 360 - 350 = 10$ Hz

45. Ans (1)

$a = \frac{50 - 40}{5} = 2 \text{ m/s}^2 \downarrow$

For 2 kg block, $20 - N = 4 \Rightarrow N = 16$ newton

46. Ans (1)

Ne maximum positive ΔHeg and He minimum ΔHeg

47. Ans (2)

Given:

E = 3.4 eV (energy of electron), $r_1 = 0.53$ Å (radius of first orbit)

Formula:

$r_n = n^2 r_1$ (Bohr radius for nth orbit), $\lambda = \frac{2\pi r_n}{n}$ (de-Broglie wavelength)

Energy of electron in hydrogen:

$E_n = -\frac{13.6}{n^2}$eV (energy levels)

Given E = 3.4 eV, the excited state corresponds to:

$E_n = -3.4 \text{ eV} \Rightarrow n^2 = \frac{13.6}{3.4} = 4 \Rightarrow n = 2$

Radius of orbit:

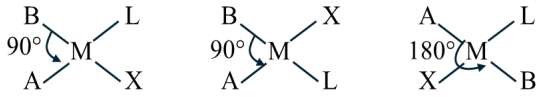
$r_n = n^2 r_1 = 2^2 \cdot 0.53 = 2.12$Å

de-Broglie wavelength:

$\lambda = \frac{2\pi r_n}{n} = \frac{2\pi \cdot 2.12}{2} \approx 6.66$Å

Final Answer: (2)

48. Ans (1)



Correct Answer : Option (1)

49. Ans (2)

$K = Ae^{-Ea/RT} \Rightarrow \log K = \log A - \frac{Ea}{2.303RT}$

$\log K_{cat.} = \log A - \frac{162 \times 10^3}{2.303\,RT}$ .....(i)

$\log K_{uncat.} = \log A - \frac{350 \times 10^3}{2.303RT}$ ....(ii)

eq^n (i) $- eq^n$ (ii)

$\log K_{cat.} - \log K_{uncat.} = \frac{(350 - 162) \times 10^3}{2.303\,RT}$

$\log \frac{K_{cat.}}{K_{uncat.}} = 32.94$

50. Ans (2)

Statement I is correct.

ψ^2 represent probability at a point inside an atom.

51. Ans (3)

Explain Question : Identify correct statement.

Concept : General properties and uses of halogens.

Solution :

(A) (True) $I_2O_5 + CO \longrightarrow I_2 + CO_2$

This reaction is used to **estimate CO** quantitatively.

(B) (True) Chlorine dioxide (ClO_2) is used for **bleaching paper and textiles**.

(C) (**False**) Actually, **Cl oxides** are **more stable** than Br oxides.

Correct order: **I > Cl > Br**.

(D) (**False**) Oxidising power decreases down the group: **Cl > Br > I**.

(E) (True) Fluorine forms **only HOF** due to **highest electronegativity and absence of d-orbitals.** Other halogens (Cl, Br, I) form multiple oxyacids.

Final Answer : Option (3)

52. Ans (2)

NCERT XII Pg. No. # 94

53. Ans (4)

C=C−Br not give SN^2

54. Ans (4)

NCERT XII Halocompound chapter

55. Ans (4)

Explaining the question : The question asks you to arrange the given transition metals in the correct order of their melting points, from highest to lowest.

Concept : Melting points depend on metallic bond strength, influenced by the number of unpaired d electrons.

Solution :

- Cr has strong bonding (high MP).
  Mn complex structure and half-filled d shell
- lead to weaker bonding (lower MP).
- Fe has intermediate strength.
  Cu filled d shell results in weaker bonding
- (lowest MP).

Order: Cr > Fe > Mn > Cu.

56. Ans (4)

BeO > MgO > CaO top to bottom acidic strength decreases

57. Ans (1)

p-block

NCERT XII Pg. No. #165

58. Ans (3)

$\Delta T_f = iK_f \times m$

$[HX \longrightarrow H^+ + X^-] \quad n = 2$

$0 - (-0.060) = [1+(2-1)\alpha] \times 1.86 \times 0.025$

$1.29 = 1 + \alpha$

$\alpha = 0.3$ (30%)

$$K_a = \frac{c\alpha^2}{(1-\alpha)} = \frac{0.025(0.3)^2}{(1-0.3)}$$

$= 3.2 \times 10^{-3}$

59. Ans (1)

A higher (more positive) standard reduction potential (E°) indicates a stronger oxidizing agent (oxidant), as the species has a greater tendency to gain electrons.

Conversely, a lower (more negative or less positive) E° indicates a stronger reducing agent (reducant) in its reduced form, as the species more easily loses electrons.

Calculation:

• Strongest Oxidant: Highest E° value is +1.52 V for MnO_4^-.

• Strongest Reductant: Lowest E° value is +0.1 V for the Sn^{4+}/Sn^{2+} couple.

reductant is the species that gets oxidized, Sn^{2+} is the strongest reductant among the choices.

Final Answer: Option(1)

60. Ans (3)

initial pH of HA

$$pH = \frac{1}{2}(pKa - \log c) = \frac{1}{2}(4 - (-1)) = 2.5$$

at half equivalence point

$$pH = pKa + \log \frac{[A^-]}{[HA]} = 4 + 0 = 4$$

at equivalence point

$$pH = 7 + \frac{1}{2}(pKa + \log c) = 7 + \frac{1}{2}(4 + \log \frac{2}{30})$$

$= 8.41$

61. Ans (1)

Electron precise means complete octet

62. Ans (3)

$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

(pink colour is due to excess of unreacted $KMnO_4$)

63. Ans (3)

NSR in Haloarene ∝ EWG

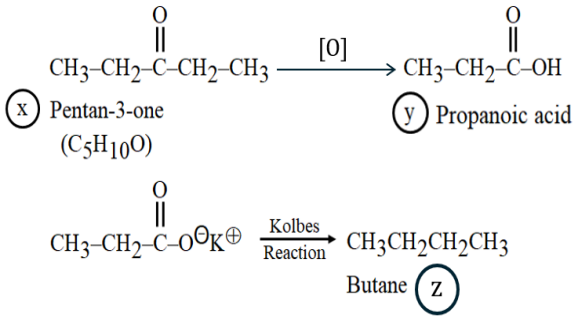
64. Ans (4)

NCERT XIth Page. No. 131

65. **Ans (1)**

Solution :

Note : Ketones doesn't gives tollen's test.



Final answer : Option (1)

66. **Ans (4)**

NCERT XII Pg. No. # 104

67. **Ans (1)**

The reactions of benzenediazonium chloride can be matched as follows:

(A) The reaction of a diazonium salt with CuCl/HCl is the Sandmeyer reaction (iv), which yields chlorobenzene.

(B) The reaction with Cu/HCl is the Gattermann reaction (iii), which also forms chlorobenzene.

(C) The reaction is the Balz-Schiemann reaction (ii), which yields fluorobenzene.

(D) The reaction with phenol in an alkaline medium is a Coupling reaction (i), which forms an azo dye.

All these reactions are nucleophilic aromatic substitutions. (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i).

Final Answer: option (1)

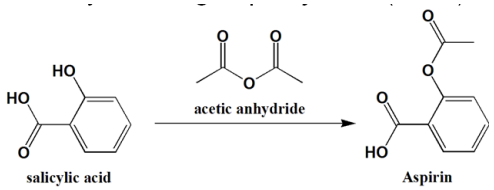
68. **Ans (4)**

Cannizzaro Reaction

69. **Ans (2)**

Solution:

Aspirin (acetylsalicylic acid) is produced by the chemical modification, or acetylation, of the hydroxyl (-OH) group in salicylic acid. Salicylic acid is the common name for o-hydroxybenzoic acid, which has a hydroxyl group and a carboxylic acid group adjacent (ortho) on a benzene ring.



Final answer: 2

70. **Ans (3)**

Answer Explanation: The reaction described is the **Carbylamine reaction** (also known as the isocyanide test). This test is specific for primary amines (both aliphatic and aromatic).

- **Primary amines** react with chloroform and an alcoholic KOH solution to form an alkyl or aryl isocyanide, which is characterized by a foul odor.
- **Secondary and tertiary amines** do not give this test. Among the given options, only Ph-NH_2 (aniline) is a primary amine. The other options are a tertiary amine ($(CH_3)_3N$), a secondary amine ($CH_3-NH-C_2H_5$), and a secondary amine ($Ph-NH-CH_3$). Therefore, only aniline will produce the foul-smelling product.

Final Answer: The correct option is option 3

71. **Ans (4)**

NCERT (2019) Page No. # 41(Theory)

72. **Ans (3)**

NCERT XI, Part-I, Pg # 176, 177 Edition 2017

For reversible process, when constant heat is given to system then entropy change is inversely proportional to the temperature.

$$\Delta S = \frac{q_{rev}}{T}$$

73. Ans (3)

A catalyst increases reaction rate by lowering activation energy but does not affect feasibility. For $\Delta G > 0$, reaction is non-spontaneous and the catalyst cannot make it proceed on its own, so assertion is incorrect. The reason is correct because the catalyst does not change Gibbs free energy change (ΔG), it only affects rate. Hence, the assertion is false but the reason is true.

Final Answer - Option (3)

74. Ans (3)

NCERT-XII, Pg. # 48, 55, 56, 57, Part-I.

75. Ans (3)

Explain Question:

Arrange ligand name in alphabetical order them write IUPAC name.

Concept:

IUPAC nomenclature, ligand name

Solution:

en ⟶ ethylenediamine

Br ⟶ bromine

pph3 ⟶ triphenylphosphine

O.N of Ni = + 2(x + 0 − 1 × 2 + 0 = 0)

Name Dibromidoethylenediaminebis (triphenyl-phosphine) nickel (II)

Final answer: Option (3)

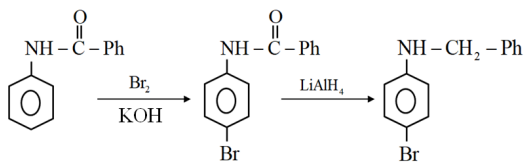
76. Ans (1)

NCERT, class 12th, part-2, Article No: 10.1.2.1 Pg No: 283, Edition : 2023 - 24.

77. Ans (1)

NAR Ald. > Ketone

78. Ans (1)



79. Ans (2)

$CH_3COONa \longrightarrow$ WASB

$$pH_{anode} = 7 + \frac{1}{2}pKa - \frac{1}{2}\log C$$

$$= 7 + \frac{1}{2}(4) + \frac{1}{2}\log\left(10^{-2}\right) = 8$$

$[H^+]$ anode $= 10^{-8}$

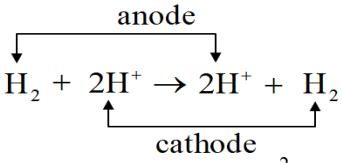
$NH_4NO_3 \longrightarrow$ SAWB

$$pH_{cathode} = 7 - \frac{1}{2}pK_b - \frac{1}{2}\log C$$

$$= 7 - \frac{1}{2}(4) - \frac{1}{2}\log(10^{-2})$$

$= 6$

$[H^+]_{cathode} = 10^{-6}$



$$E = 0 - \frac{0.06}{2}\log\frac{\left(10^{-8}\right)^2}{\left(10^{-6}\right)^2}$$

$= -0.03\ (-4) = 0.12$ volt

80. Ans (2)

Explain Question: Asking about matching complexes with their number of unpaired electrons.

Concept: Crystal Field Theory, high-spin vs low-spin

Solution:

• $[CrF_6]^{4-} \longrightarrow Cr^{2+}$, d^4, weak-field (F) ⟶ high-spin ⟶ 4 unpaired ⟶ matches S

• $[MnF_6]^{4-} \longrightarrow Mn^{2+}$, d^5, weak-field (F) ⟶ high-spin ⟶ 5 unpaired ⟶ matches P

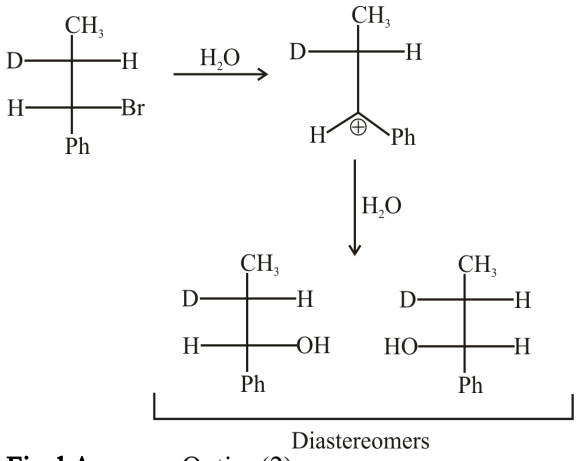
• $[Cr(CN)_6]^{4-} \longrightarrow Cr^{2+}$, d^4, strong-field (CN) ⟶ low-spin ⟶ 2 unpaired ⟶ matches Q

• $[Mn(CN)_6]^{4-} \longrightarrow Mn^{2+}$, d^5, strong-field (CN) ⟶ low-spin ⟶ 1 unpaired ⟶ matches R

Final answer: Option (2)

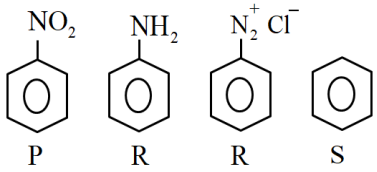
81. Ans (2)

Solution:



Final Answer : Option(2)

82. Ans (2)



83. Ans (4)

$XCO_3(s) \rightleftharpoons XO(s) + CO_2(g)$

t = 0 ⇒ 4 - 0

Eq. ⇒ (4−x) x x

$K_p = P_{CO_2} = 1.642$ atm, T = 727°C = 1000 K

$1.642 = \left(\frac{x}{50}\right) \times (0.0821 \times 1000)$

x = 1

% reacted = $\frac{1}{4} \times 100 = 25\%$

% unreacted = 75%

84. Ans (4)

$C_2H_5OH(l) \xrightarrow{\Delta H} CH_3OCH_3(g)$

$\downarrow 41$ $\qquad$ $\uparrow -6 \times 415 - 2 \times 352 = -3194$

$C_2H_5OH(g) \longrightarrow 2C(g) + 6H(g) + O(g)$

348 + 352 + 5 × 415 + 463

= 3238

$\Delta rH = 3238 + 41 - 3194$

= 85 kJ / mole

85. Ans (4)

$X_2O_3 + 3H_2 \longrightarrow 3H_2O + 2X$

$\downarrow \qquad \downarrow$

$n = \frac{0.1596}{2x + 48} \quad \frac{6 \times 10^{-3}}{2}$

1 mol of X_2O_3 is reduced by = 3 mol H_2

So $\frac{0.1596}{2x + 48}$ X_2O_3 is reduced by

$= \frac{3 \times 0.1596}{2x + 48} = \frac{6 \times 10^{-3}}{2}$

$(2x + 48) \times 10^{-3} = 0.1596$

2x + 48 = 159.6

2x = 159.6 − 48

2x = 111.6

$x = \frac{111.6}{2} = 55.8$

86. Ans (4)

Question Explanation:

Basic salt used in Borax bead test is?

Concept:

Borax Beat test.

Solution:

For borax beat test the salt is required is $Na_2B_4O_7 \cdot 10H_2O$.

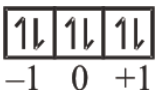
$Na_2B_4O_7 \cdot 10H_2O \xrightarrow{\Delta} Na_2B_4O_7 \xrightarrow{\Delta} \underbrace{2NaBO_2 + B_2O_3}_{\text{Glassy bead}}$

It is used to test d-block cations.

Final Answer: Option(4)

87. Ans (4)

(a) Mg = $1s^2$ $2s^2$ $2p^6$ $3s^2$
ℓ = 0 0 1 0
m = 0 0 1 0 for m = 0, No. of e^- = 8



(b) One orbital occupied maximum $2e^-$

(c) M shell n = 3, no. of $e^- = 2n^2 = 2 \times 3^2 = 18$

(d) $P = 1s^2 2s^2 2p^6 3s^2 3p^3$

l = 0 for s orbital, so $6e^-$, in s, so for 3e, $s = +\frac{1}{2}$

and for 3e, $s = -\frac{1}{2}$

88. Ans (3)

$CH_3 - CN \xrightarrow{H_3O^+} CH_3 - COOH \xrightarrow{LiAlH_4} CH_3 - CH_2 - OH$

89. Ans (4)

NCERT XII, Biomolecules chapter

90. Ans (3)

Explain the question : Identify which is not correctly matched.

Concept : This question is based on tollen's, fehlings, and iodoforms test.

Solution : Methyl ketone ($CH_3-\overset{\overset{O}{\|}}{C}-$) containing compounds can give iodoform test.

Ex. $CH_3-\overset{\overset{O}{\|}}{C}-H$ and $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

Final answer : 3

91. Ans (2)

NCERT XI, Pg. # 8

92. Ans (4)

NCERT XI, Pg. # 14,15

93. Ans (2)

NCERT-XI, Page # 20

94. Ans (4)

NCERT XI, Pg. # 18

95. Ans (3)

NCERT-XI Pg # 26, 27

96. Ans (1)

NCERT XI, Pg. # 29

97. Ans (3)

NCERT Pg.#30-32

98. Ans (4)

Module-2, NCERT XI

99. Ans (1)

NCERT-XII, Pg. No. # 61

100. Ans (4)

NCERT Page No. # 66,67

101. Ans (2)

NCERT Pg # 75

102. Ans (1)

NCERT, Pg. # 76,77

103. Ans (2)

NCERT-XII, Pg. No. # 18, 19

104. Ans (4)

NCERT XII, Pg. # 22

105. Ans (3)

NCERT XII, Pg. # 10

106. Ans (4)

NCERT Pg.# 206

107. Ans (4)

NCERT (XII) ; Page No. # 199

108. Ans (3)

NCERT XII, Pg. # 196, 197, 198

109. Ans (3)

NCERT Pg. # 206

110. Ans (4)

NCERT Pg. # 213

111. Ans (3)

NCERT Pg. # 205 & 210

112. Ans (2)

NCERT Pg. # 223, Hindi Pg. # 243

113. **Ans (4)**

Question Explanation: The question asks to find the correct statement.

Concept : Biodiversity And Conservation

Explanation :

All the biodiversity hotspots put together more than 50 percent of earth's land area. - Incorrect

All the biodiversity hotspots put together cover less than 2 percent of the earth's

land area,

Final answer : All the biodiversity hotspots put together more than 50 percent of earth's land area.

114. **Ans (3)**

Explain Question : The question asks about the examples of the ex-situ conservation in the given options.

Concept : This question is based on Ex-situ Conservation.

Solution :

- The list of ex situ conservations are:
- Zoological parks.
- Botanical gardens.
- Cryopreservation.
- Wildlife safari. (Total : 4)

So, the correct answer is: 3.

Final answer : Option (3)

115. **Ans (2)**

NCERT-XII, Pg. # 78

116. **Ans (1)**

NCERT-XII, Pg. # 71

117. **Ans (3)**

NCERT XII, Page No. # 76

118. **Ans (3)**

NCERT, Pg # 74,75,76

119. **Ans (4)**

NCERT, Pg. # 63-64

120. **Ans (1)**

NCERT-XII Pg # 89

121. **Ans (3)**

NCERT XII Pg # 108

122. **Ans (1)**

NCERT-XII, Pg. No. # 98

123. **Ans (2)**

NCERT-XII, Pg. No. # 73, 74, 75

124. **Ans (2)**

Find the codon on mRNA and find the respective anti-codon on tRNA from 3' $\longrightarrow$ 5' direction.

125. **Ans (2)**

NCERT-XII Pg.# 85, 94, 96, 101

126. **Ans (1)**

NCERT, Pg. # 174

127. **Ans (3)**

NCERT, Pg. # 169

128. **Ans (4)**

NCERT Pg. # 168, 169

129. **Ans (2)**

NCERT-XII, Pg. # 179, 180 (E), 196, 197 (H)

130. **Ans (1)**

NCERT-XII, Pg. # 172-174

131. **Ans (2)**

NCERT-XII, Pg. # 167

132. **Ans (2)**

NCERT-XIIth, Pg. # 157

133. **Ans (4)**

NCERT-XII, Pg. # 151, 153 (E), 166, 168 (H)

134. **Ans (2)**

NCERT XII Page No. # 150

135. **Ans (2)**

NCERT, Pg. # 88,89,90

136. **Ans (3)**

NCERT, Pg. # 96, 97

137. **Ans (2)**

NCERT-XI, Pg.# 99

138. Ans (3)
1 Bivalent have
2 chromosome, 2 centromere and 4 kinetochores

139. Ans (1)
NCERT, Pg. 126-127

140. Ans (2)
NCERT XI Pg. 126

141. Ans (4)
NCERT XI Pg. # 109, 110

142. Ans (2)
NCERT XI Pg.# 86

143. Ans (1)
NCERT XI Pg. # 106

144. Ans (3)
NCERT XI Pg.# 118

145. Ans (3)
NCERT XI Pg. # 143

146. Ans (2)
NCERT-XI, Pg. # 149-150

147. Ans (1)
NCERT XI Pg. # 156

148. Ans (4)
NCERT XI Page # 156-159

149. Ans (3)
NCERT XI Pg. # 168, 169

150. Ans (3)
NCERT XI Pg. # 175, 177

151. Ans (4)
NCERT-XI Pg. # 175 - 177

152. Ans (4)
NCERT Pg# Eng.(154) H(167)

153. Ans (2)
NCERT 12th Pg. No. 146

154. Ans (3)
NCERT-XII Pg # 151, 152 (Option 4 from Module)

155. Ans (3)
NCERT XII Page No. # 157

156. Ans (2)
NCERT Pg. # 114

157. Ans (2)
NCERT Pg. # 119

158. Ans (3)
NCERT-XIIth, Page#51, 52, 53

159. Ans (4)
NCERT Pg. # 53

160. Ans (4)
NCERT Pg. # 47-48

161. Ans (1)
NCERT Pg. # 64

162. Ans (3)
NCERT Pg. # 51

163. Ans (1)
NCERT XI Pg. # 195

164. Ans (2)
NCERT Pg. # 201

165. Ans (1)
NCERT Pg. 189

166. Ans (3)
NCERT Pg. # 183-184

167. Ans (2)
NCERT Pg # 241,242,245,246

168. Ans (1)
NCERT Pg. # 246

169. Ans (3)
NCERT XI Pg. 245-247

170. Ans (1)
NCERT XI Pg. 231

171. Ans (4)
NCERT Pg. # 322(E), 321(H)

172. Ans (2)
NCERT Pg. # 205

173. Ans (4)
NCERT Pg. # 209

174. **Ans (1)**
NCERT Pg.#226.

175. **Ans (4)**
NCERT Pg. # 101

176. **Ans (3)**
Old NCERT_Pg. No. - 118

177. **Ans (3)**
NCERT Pg.# XI- 113, 114.

178. **Ans (3)**
NCERT Pg. # 47

179. **Ans (2)**
NCERT Pg.#45.

180. **Ans (1)**
NCERT Pg. # 45